

Week 3 Installation Guide

Enterprise Server Deployment and Operating System Installation

Installation Guide for Ubuntu Server 26.04 LTS in a Virtual Machine

Week 3 Portfolio Project, ITEP 414, System Administration and Maintenance

Prepared by: Symon Kiel G. Beato

Instructor: John Randolph M. Penaredondo, MIT

1. Project Overview

Operating systems serve as the foundation of every enterprise IT infrastructure. Before a server can host websites, databases, cloud services, or enterprise applications, it must first be properly installed, configured, secured, and documented. In this project, I acted as a Junior System Administrator of ABC Startup Solutions, a startup company deploying its first Linux server. I performed a complete Ubuntu Server installation in a virtual machine, configured the essential settings, verified the installation, compared modern boot technologies, and documented every step so that future administrators can reproduce the installation.

This guide records the exact procedure followed, the configuration values used, the verification results, and the problems encountered during deployment. All screenshots were captured from the actual virtual machine named Ubuntu-Server-Week03.

2. Objectives

- Install Ubuntu Server 26.04 LTS in a virtual machine using enterprise recommended practices.
- Configure server settings such as hostname, user account, disk partitioning, and SSH.
- Verify the installation through login, network, and service checks.
- Compare BIOS and UEFI firmware and explain the boot process.
- Compare Ubuntu Server with Windows Server and Rocky Linux.
- Produce professional documentation suitable for future administrators.

3. Software Requirements

Software	Version	Purpose
Oracle VirtualBox	7.x (latest stable)	Virtualization platform for the guest server
Ubuntu Server ISO	26.04 LTS	Operating system installer for the server
Windows Server Evaluation ISO	Latest evaluation release	Bring-home activity (comparison install)

4. Virtual Machine Specifications

Component	Configuration
Name	Ubuntu-Server-Week03
Type / Version	Linux, Ubuntu (64-bit)
RAM	4096 MB (4 GB)
CPU	2 virtual processors
Storage	40 GB virtual disk (VDI, dynamically allocated)
Network	NAT (default adapter)
Optical Drive	ubuntu-26.04-live-server-amd64.iso (removed after installation)

5. Step-by-Step Installation Procedure

5.1 Create the Virtual Machine

1. Open Oracle VirtualBox and click **New**.
2. Set the name to Ubuntu-Server-Week03, type **Linux**, version **Ubuntu (64-bit)**.
3. Set the base memory to **4096 MB** and the number of CPUs to **2**.
4. Create a virtual hard disk of **40 GB** (VDI format).
5. Open the virtual machine settings and attach the Ubuntu Server ISO to the optical drive.
6. Confirm the network adapter is set to **NAT**, then start the machine.

The wizard shows the hardware summary before the machine is created. Verify that memory, CPU, and disk match the specifications before clicking Finish.

5.2 Run the Installer

1. At the first menu, choose **Try or Install Ubuntu Server** and press Enter. The installer loads the kernel and the installer environment, which takes a few minutes.
2. Select **English** as the language.
3. Select **English (US)** as the keyboard layout.
4. On the network screen, accept the default DHCP configuration. The installer assigns the address 10.0.2.15 to the adapter enp0s3; record this address for later SSH use.
5. Set the hostname to server01.
6. Create the administrative user: full name **Symon Kiel G. Beato**, username beato, with a strong password.
7. For disk setup, choose **Guided, use entire disk** and confirm the partition layout.
8. On the SSH setup screen, enable **Install OpenSSH server**. In this installation the option was skipped, so SSH was installed manually afterward (see Section 11).
9. Do not install any additional packages.
10. Wait for the installation to finish, then reboot when prompted and remove the installation medium.

```

Ubuntu 26.04 LTS server01 tty1

server01 login: beato
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Fri Aug 14 07:24:10 AM UTC 2026

System load:            0.02
Usage of /:              10.4% of 39.07GB
Memory usage:           11%
Swap usage:             0%
Processes:              121
Users logged in:        0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fed8:dc21

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```

Figure 1. Login to the newly installed server. The welcome message confirms Ubuntu 26.04 LTS and the server01 hostname.

6. Configuration Details

Setting	Value
Hostname	server01
User account	beato (administrative, sudo)
IP address (DHCP)	10.0.2.15
Partition scheme	Guided, entire disk (single LVM volume group)
File system	ext4 (root filesystem, 39.07 GB usable)
Disk size	40 GB
SSH service	openssh-server, enabled and running on port 22
Package sources	http://archive.ubuntu.com/ubuntu (switched from the Philippines mirror)

7. Server Verification

Task 1: Login

Log in using the account created during installation. A successful login shows the message of the day, the system statistics, and the prompt `beato@server01:~$`.

Task 2: Verify Hostname

Run the hostname command. The output must be server01, which matches the hostname configured during installation.

```
beato@server01:~$ hostname
server01
```

Figure 2. The hostname command returns server01.

Task 3: Verify IP Address

Run the ip addr command and locate the enp0s3 adapter. The assigned IPv4 address is 10.0.2.15, which is the standard NAT address issued by VirtualBox.

```
beato@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d8:dc:21 brd ff:ff:ff:ff:ff:ff
    altname enx080027d8dc21
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85535sec preferred_lft 85535sec
    inet6 fd17:625c:f037:2:a00:27ff:fed8:dc21/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86124sec preferred_lft 14124sec
    inet6 fe80::a00:27ff:fed8:dc21/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Figure 3. The ip addr output shows inet 10.0.2.15 on adapter enp0s3.

Task 4: Test Internet Connectivity

Run ping -c 4 google.com. Four replies with zero packet loss confirm that the server can reach the internet through the NAT network.

```
beato@server01:~$ ping -c 4 google.com
PING google.com (142.250.198.206) 56(84) bytes of data.
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=1 ttl=64 time=36.7 ms
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=2 ttl=64 time=43.1 ms
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=3 ttl=64 time=39.8 ms
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=4 ttl=64 time=38.8 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3096ms
rtt min/avg/max/mdev = 36.718/39.613/43.096/2.300 ms
```

Figure 4. Ping to google.com succeeds with 0% packet loss.

Task 5: Update the Server

Run sudo apt update to refresh the package lists, then sudo apt upgrade -y to install the available updates. In this deployment, 51 packages were upgraded. Updating the server immediately after installation is a mandatory step for security and stability.

```

Summary:
  Upgrading: 51, Installing: 0, Removing: 0, Not Upgrading: 2
  Download size: 717 MB
  Space needed: 1,174 kB / 36.2 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 motd-news-config all 14ubuntu6.1 [5,632 B]
Get:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu6.1 [75.0 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcrypt20 amd64 1.12.0-2ubuntu1 [671 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-1ubuntu1.1 [1,198 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libapparmor1 amd64 5.0.2-0ubuntu1~26.04.1 [51.3 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-minimal amd64 1.570.2 [8,594 B]
Get:7 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apparmor amd64 5.0.2-0ubuntu1~26.04.1 [380 kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-host amd64 1:9.20.24-1ubuntu0.1 [49.8 kB]
Get:9 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-dnsutils amd64 1:9.20.24-1ubuntu0.1 [164 kB]
Get:10 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-libs amd64 1:9.20.24-1ubuntu0.1 [1,308 kB]
Get:11 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-core all 1:26.04.22 [24.9 kB]
Get:12 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-distupgrade all 1:26.04.22 [102 kB]
Get:13 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-standard amd64 1.570.2 [8,582 B]
Get:14 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-problem-report all 2.34.1-0ubuntu0.1 [17.4 kB]
Get:15 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apport all 2.34.1-0ubuntu0.1 [36.4 kB]
Get:16 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport-core-dump-handler all 2.34.1-0ubuntu0.1 [9,674 B]
Get:17 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport all 2.34.1-0ubuntu0.1 [83.9 kB]
Get:18 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init-base all 26.1-0ubuntu3~26.04.1 [587 kB]
Get:19 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init all 26.1-0ubuntu3~26.04.1 [2,119 B]
Get:20 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 fwupd amd64 2.1.1-1ubuntu3.1 [12.7 MB]
Get:21 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfwupd3 amd64 2.1.1-1ubuntu3.1 [145 kB]
Get:22 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpackagekit-glib2-18 amd64 1.3.4-3ubuntu1.1 [121 kB]
Get:23 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-packagekitglib-1.0 amd64 1.3.4-3ubuntu1.1 [25.5 kB]
Get:24 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub-pc amd64 2.14-2ubuntu2.1 [124 kB]
Get:25 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub-pc-bin amd64 2.14-2ubuntu2.1 [1,325 kB]
Get:26 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub2-common amd64 2.14-2ubuntu2.1 [2,586 kB]
Get:27 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsgutils2-1.48 amd64 1.48-3ubuntu3.1 [120 kB]
Err:28 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-amd-misc all 20260319.git217ca6e4-0ubuntu1.1
403 Forbidden [IP: 202.79.180.254 80]
Get:29 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-amd-graphics all 20260319.git217ca6e4-0ubuntu3.1 [29.6 MB]
Get:30 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-broadcom-wireless all 20260319.git217ca6e4-0ubuntu1.1 [11.2 MB]
Get:31 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-intel-graphics all 20260319.git217ca6e4-0ubuntu2.1 [22.0 MB]
Get:32 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-intel-misc all 20260319.git217ca6e4-0ubuntu1.2 [10.1 MB]
Get:33 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-intel-wireless all 20260319.git217ca6e4-0ubuntu2.1 [102 MB]
Get:34 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-marvell-prestera all 20260319.git217ca6e4-0ubuntu1.2 [32.3 MB]
Get:35 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-marvell-wireless all 20260319.git217ca6e4-0ubuntu1.1 [7,434 kB]
Get:36 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-mediatek all 20260319.git217ca6e4-0ubuntu1.2 [27.9 MB]
Get:37 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-mellanox-spectrum all 20260319.git217ca6e4-0ubuntu1.1 [61.9 MB]
Get:38 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-misc all 20260319.git217ca6e4-0ubuntu2.2 [31.9 MB]
Get:39 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-netronome all 20260319.git217ca6e4-0ubuntu1.1 [5,765 kB]
Get:40 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-nvidia-graphics all 20260319.git217ca6e4-0ubuntu1.1 [109 MB]
Get:41 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-qlogic all 20260319.git217ca6e4-0ubuntu1.1 [7,178 kB]
Get:42 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-qualcomm-graphics all 20260319.git217ca6e4-0ubuntu1.1 [11.0 MB]
Get:43 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-qualcomm-misc all 20260319.git217ca6e4-0ubuntu2.1 [167 MB]
81% [43] linux-firmware-qualcomm-misc 86.2 MB/167 MB 52%] 17.4 MB/s B/s

```

Figure 5. apt upgrade downloading and installing the available packages.

```

beato@server01:~$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14          pollinate
Use 'sudo apt autoremove' to remove them.

Upgrading:
  apparmor                grub-pc                linux-firmware-broadcom-wireless  linux-firmware-qlogic
  apport                  grub-pc-bin            linux-firmware-intel-graphics     linux-firmware-qualcomm-graphics
  apport-core-dump-handler  grub2-common          linux-firmware-intel-misc         linux-firmware-qualcomm-misc
  base-files              iproute2               linux-firmware-intel-wireless     linux-firmware-qualcomm-wireless
  bind9-dnsutils          libapparmor1           linux-firmware-marvell-prestera   linux-firmware-realtek
  bind9-host              libfwupd3              linux-firmware-marvell-wireless   motd-news-config
  bind9-libs              libgcrypt20            linux-firmware-mediatek           packagekit
  cloud-init              libpackagekit-glib2-18  linux-firmware-mellanox-spectrum  python3-apport
  cloud-init-base         libsgutils2-1.48       linux-firmware-misc               python3-distupgrade
  fwupd                   linux-firmware-amd-graphics  linux-firmware-netronome          python3-problem-report
  gir1.2-packagekitglib-1.0  linux-firmware-amd-misc  linux-firmware-nvidia-graphics    sg3-utils

Not upgrading yet due to phasing:
  python3-software-properties  software-properties-common

Summary:
  Upgrading: 51, Installing: 0, Removing: 0, Not Upgrading: 2
  Download size: 584 kB / 717 MB
  Space needed: 1,174 kB / 35.5 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-amd-misc all 20260319.git217ca6e4-0ubuntu1.1
Fetched 584 kB in 2s (242 kB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
(Reading database... 136625 files and directories currently installed.)
Preparing to unpack ../motd-news-config_14ubuntu6.1_all.deb...
Unpacking motd-news-config (14ubuntu6.1) over (14ubuntu6)...
Preparing to unpack ../base-files_14ubuntu6.1_amd64.deb...
Unpacking base-files (14ubuntu6.1) over (14ubuntu6)...

```

Figure 6. Upgrade progress after the package sources were fixed (see Section 11).

Task 6: Verify SSH Service

Run `systemctl status ssh`. The service must display **active (running)** and be enabled, meaning it starts automatically on every boot and listens on port 22.

```
beato@server01:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 07:21:26 UTC; 7s ago
   Invocation: 8baa47842a2549c88d23afdf028bdcd2
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 14436 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 14439 (sshd)
     Tasks: 1 (limit: 3992)
    Memory: 1.2M (peak: 2.2M)
       CPU: 41ms
   CGroup: /system.slice/ssh.service
           └─14439 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 07:21:26 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 07:21:26 server01 sshd[14439]: Server listening on 0.0.0.0 port 22.
Aug 14 07:21:26 server01 sshd[14439]: Server listening on :: port 22.
Aug 14 07:21:26 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Figure 7. The ssh service is active (running), enabled, and listening on port 22.

8. BIOS vs UEFI Comparison

Firmware is the first software that runs when a computer is powered on. BIOS is the legacy firmware that reads the Master Boot Record of a disk, while UEFI is the modern firmware that loads boot loaders from a GPT disk and adds security features such as Secure Boot. The table below summarizes the main differences.

Aspect	BIOS	UEFI
Boot process	Reads the MBR (first 512 bytes of the disk)	Reads .efi boot loaders from the EFI System Partition
Maximum disk	2 TB	Up to 9.4 ZB (GPT)
Partition style	MBR (max 4 primary partitions)	GPT (up to 128 partitions)
Security	None	Secure Boot verifies boot loader signatures
Speed	Slower, sequential initialization	Faster, parallel initialization
Modern usage	Legacy and compatibility only	Standard on all new computers

UEFI has largely replaced BIOS because it removes the 2 TB disk limit, boots faster, and adds Secure Boot protection against bootkits. Microsoft required UEFI for Windows 8 certification in 2012, which pushed every hardware vendor to adopt it. The full comparison is documented in the separate file BIOS_vs_UEFI.pdf.

9. Boot Process Flowchart

The boot process of the server follows eight stages, from power-on to the login prompt. The flowchart below was created with draw.io and is also available as PNG and PDF in the diagrams folder.

Ubuntu Server Boot Process

Flowchart of the startup sequence from power-on to login prompt

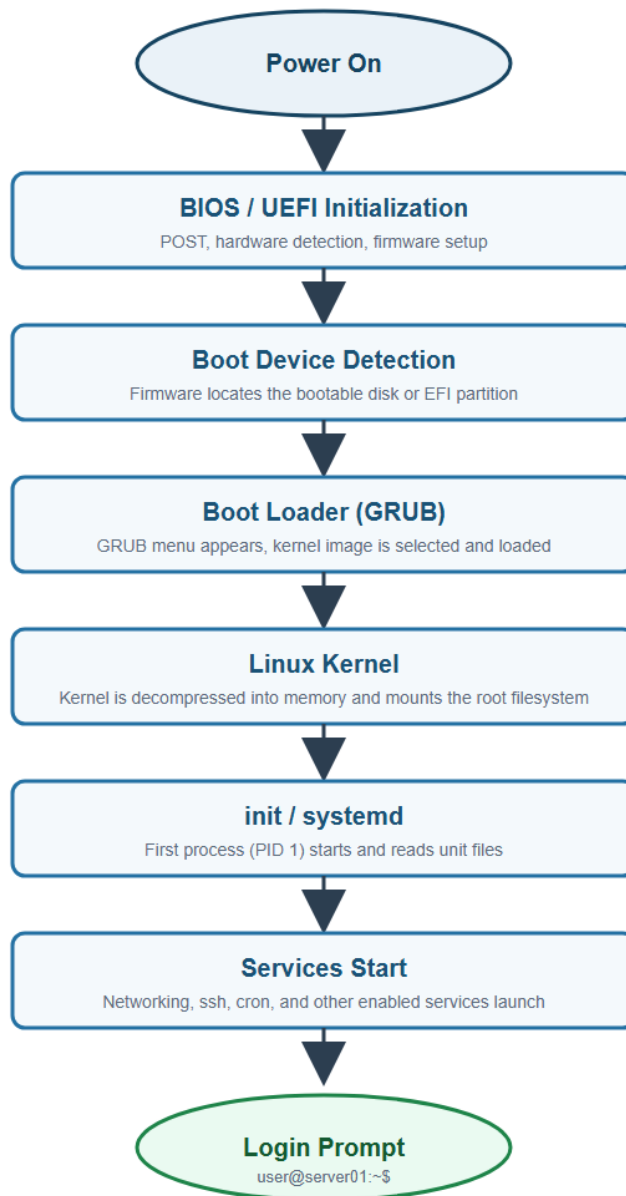


Figure 8. Ubuntu Server boot process: Power On, BIOS/UEFI initialization, boot device detection, GRUB, kernel, systemd, services, login prompt.

10. Windows Server Installation Summary (Bring-Home Activity)

As part of the bring-home assignment, Windows Server Evaluation was installed in a separate virtual machine to compare the installation experience with Ubuntu Server. The minimum tasks were: install the operating system, assign a computer name, create an administrator password, and log in successfully.

Windows Server uses a graphical installer with wizard pages, which is very different from the text-based installer of Ubuntu Server. The version installed was **Windows Server 2025 Standard Evaluation with Desktop Experience**, which includes the full graphical environment. The evaluation edition does not require a product key and is licensed for 180 days. The About Windows dialog confirmed the installation: Windows Server 2025 Standard Evaluation, Version 24H2, OS Build 26100.32230.

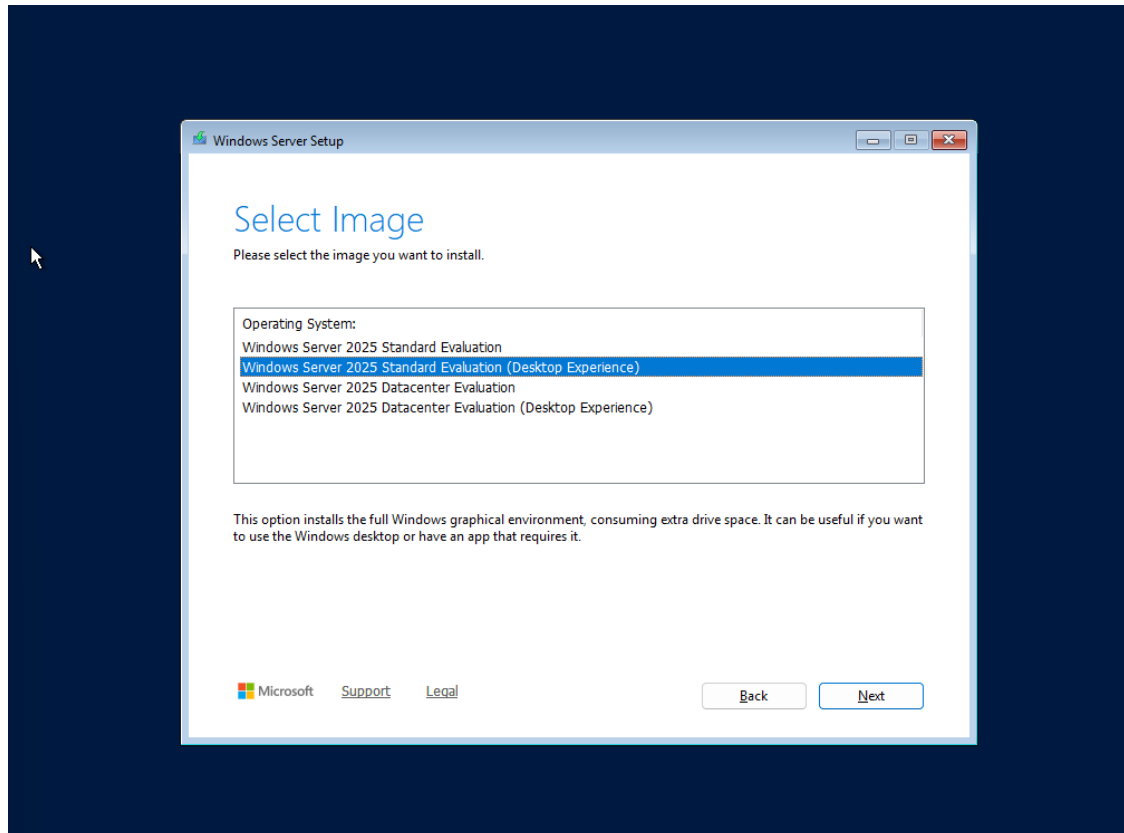


Figure 9. Selecting the Windows Server 2025 Standard Evaluation (Desktop Experience) image during setup.

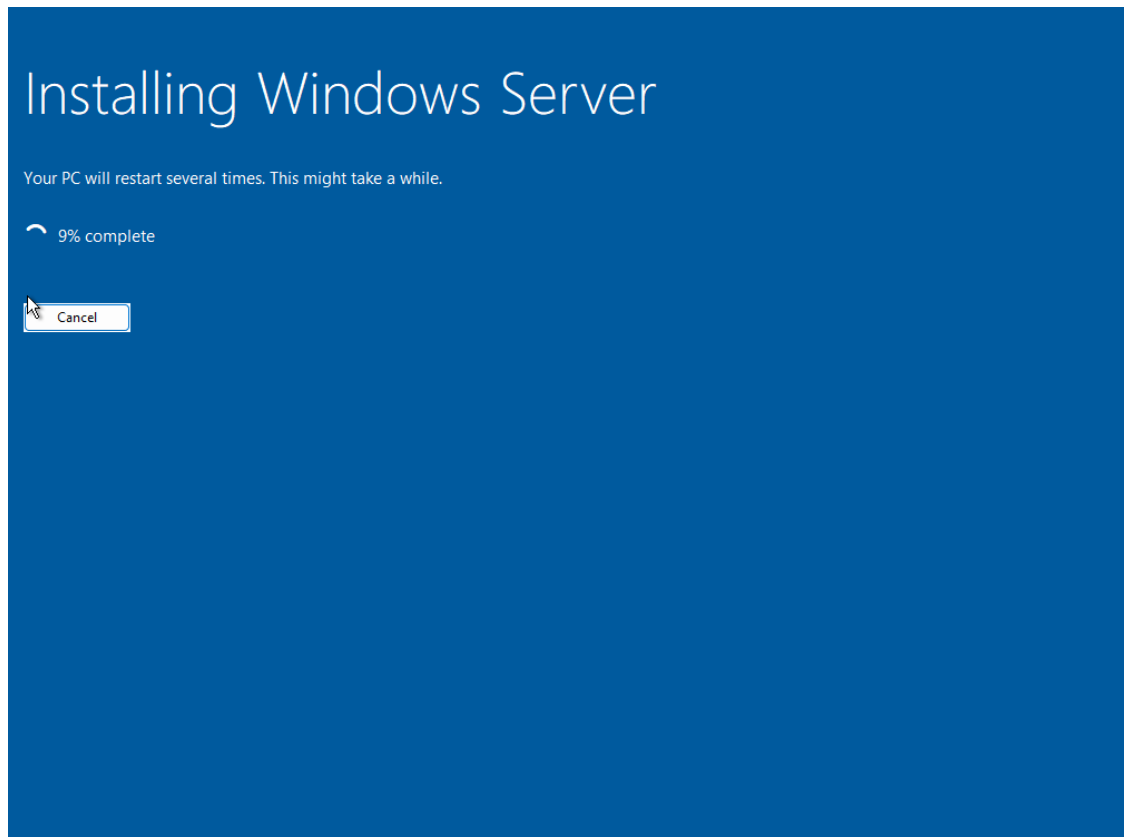


Figure 10. The Windows Server installation progress screen.

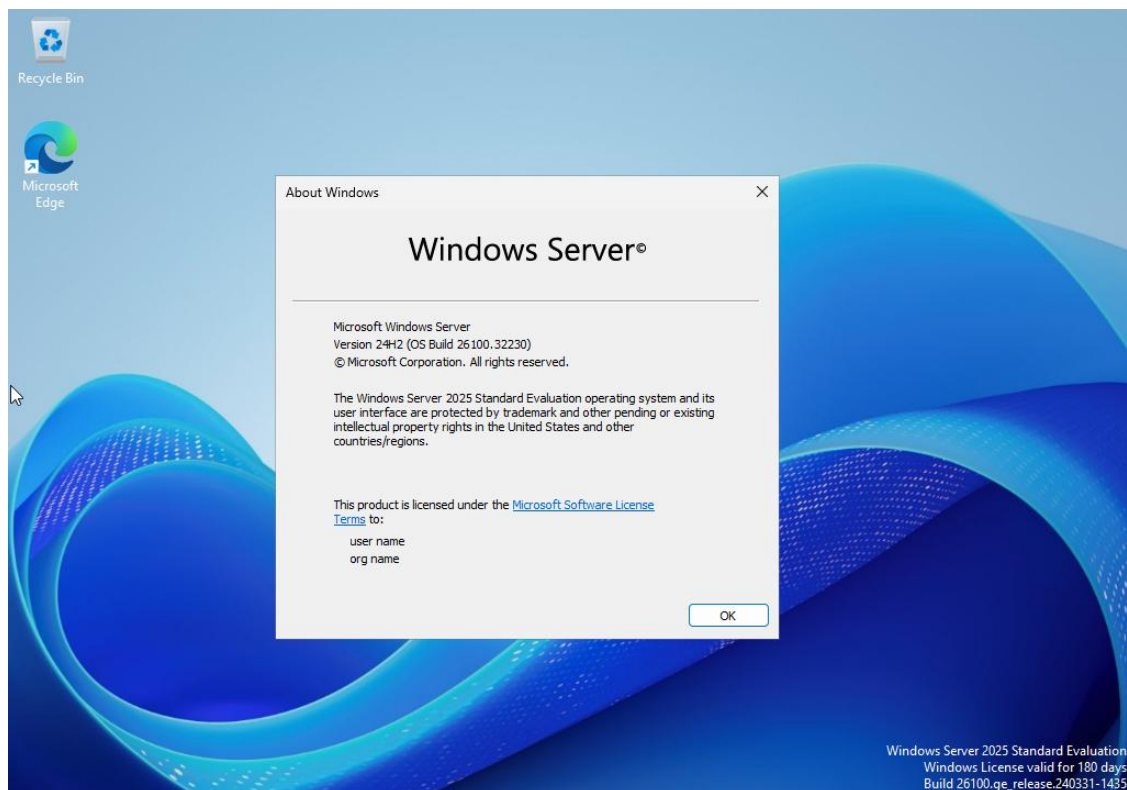


Figure 11. The About Windows dialog confirming the installed version: Windows Server 2025 Standard Evaluation, Version 24H2.

After installation, the server requested an administrator password, and logging in opened the Server Manager console, which is the central management tool of Windows Server. The installation experience confirmed the main differences between the two platforms: Windows Server relies on a wizard-driven graphical setup and license-based editions, while Ubuntu Server uses a text-based installer and a free licensing model. The screenshots are also collected in the screenshots folder of this repository.

11. Operating System Comparison Report

Enterprises typically choose between three families of server operating systems: Windows Server, Ubuntu Server, and Rocky Linux. Each one has a different licensing model, management style, and ecosystem. The table below compares them across the criteria that matter most to administrators.

Criterion	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary, per-core subscription or CALs	Free and open source	Free and open source (RHEL rebuild)
User interface	Graphical (Server Manager, PowerShell)	Command line by default	Command line by default
Package management	MSI, MSIX, Windows Update	APT (deb packages)	DNF (rpm packages)
Security	BitLocker, Defender, Active Directory	AppArmor, UFW, regular patching	SELinux, firewalld, NSA-grade hardening
Performance	Heavier footprint, GUI overhead	Lightweight, efficient on low RAM	Very stable, tuned for production workloads
Typical use cases	Active Directory, Exchange, .NET apps	Web hosting, databases, cloud workloads	Web servers, virtualization hosts, enterprise apps
Advantages	Enterprise support, integration with Microsoft tools	Free, huge community, easy updates	Free RHEL compatibility, excellent stability
Disadvantages	Costly licenses, heavy resource use	No official long-term paid support by default	Smaller community, fewer cloud vendor integrations

Windows Server is the strongest choice for organizations that are already invested in the Microsoft ecosystem, because it offers Active Directory, Group Policy, and seamless integration with Office 365 and Azure. The cost, however, is significant, and the operating system demands more memory and disk than its Linux counterparts.

Ubuntu Server is ideal for startups and cloud deployments. It is free, receives regular security updates, and is the most common guest operating system on public clouds. The package manager APT makes installing and updating software straightforward, and the large community means that almost every problem has a documented answer.

Rocky Linux is the community rebuild of Red Hat Enterprise Linux. It shares RHEL's stability and SELinux hardening but costs nothing, which makes it popular for production servers that need enterprise grade reliability without the subscription fee. Its DNF package manager and SELinux default differ from Ubuntu, so administrators need some adaptation.

For ABC Startup Solutions, Ubuntu Server was the right choice for the first deployment: it is free, lightweight, well documented, and fully sufficient for file sharing, database hosting, and web hosting.

12. Problems Encountered and Solutions

Problem	Solution
The OpenSSH server was not installed because the option was skipped during installation.	Installed it manually with <code>sudo apt install openssh-server -y</code> , then enabled and started it with <code>sudo systemctl enable --now ssh</code> .
apt upgrade failed with HTTP 403 Forbidden from the <code>ph.archive.ubuntu.com</code> mirror.	Changed the repository URIs to <code>archive.ubuntu.com</code> in the file <code>/etc/apt/sources.list.d/ubuntu.sources</code> , then ran <code>sudo apt update</code> again.
After editing the sources file, apt reported "Malformed entry 1 ... (URI)".	Reopened the file in nano, corrected the two URI lines to include the <code>/ubuntu</code> path, saved, and ran <code>sudo apt update</code> successfully.
The virtual machine window became unresponsive during the upgrade.	Restarted the virtual machine and repaired the interrupted installation with <code>sudo dpkg --configure -a</code> followed by <code>sudo apt -fix-broken install -y</code> .
The password was typed visibly during the first login attempt, exposing it.	Changed the password immediately with the <code>passwd</code> command and recorded the new password in a secure location.

13. Lessons Learned

- Always enable OpenSSH during installation if remote administration is planned; installing it later is possible but requires an extra step and a console session.
- Mirror servers can fail intermittently. A 403 Forbidden error is usually fixed by retrying, and permanently by switching to the main archive mirror.
- After editing configuration files, always test with the corresponding command (such as `sudo apt update`) to catch syntax errors immediately.
- Interrupted package operations must be repaired with `dpkg --configure -a` and `apt --fix-broken install` before continuing.
- Recording the IP address during installation saves time when connecting over SSH later.

14. References

- Ubuntu Server Documentation, <https://ubuntu.com/server/docs>
- Ubuntu Installation Guide, <https://help.ubuntu.com/community/Installation>
- Oracle VirtualBox User Manual, <https://www.virtualbox.org/manual>
- Microsoft Evaluation Center, Windows Server, <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2025>
- Rocky Linux Documentation, <https://docs.rockylinux.org>
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